

REVERSE ENGINEERING REPORT: AI POWERED PERSONALISED CAREER RECOMMENDATION AND SKILL ROADMAP GENERATION SYSTEM

Course Code & Name:	CSA17 – Artificial Intelligence
Course Outcome:	CO5 – Design AI-based systems using PySpark, RDD operations, and intelligent agent architectures (BTL 5)
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Target Architecture:	Distributed Scalable AI System Architecture

1. System Decomposition & Analysis

System Concept: The target application is an AI Powered Personalised Career Recommendation and Skill Roadmap Generation System designed to analyze academic backgrounds, skills, interests, and career goals to produce tailored career paths and structured learning roadmaps.

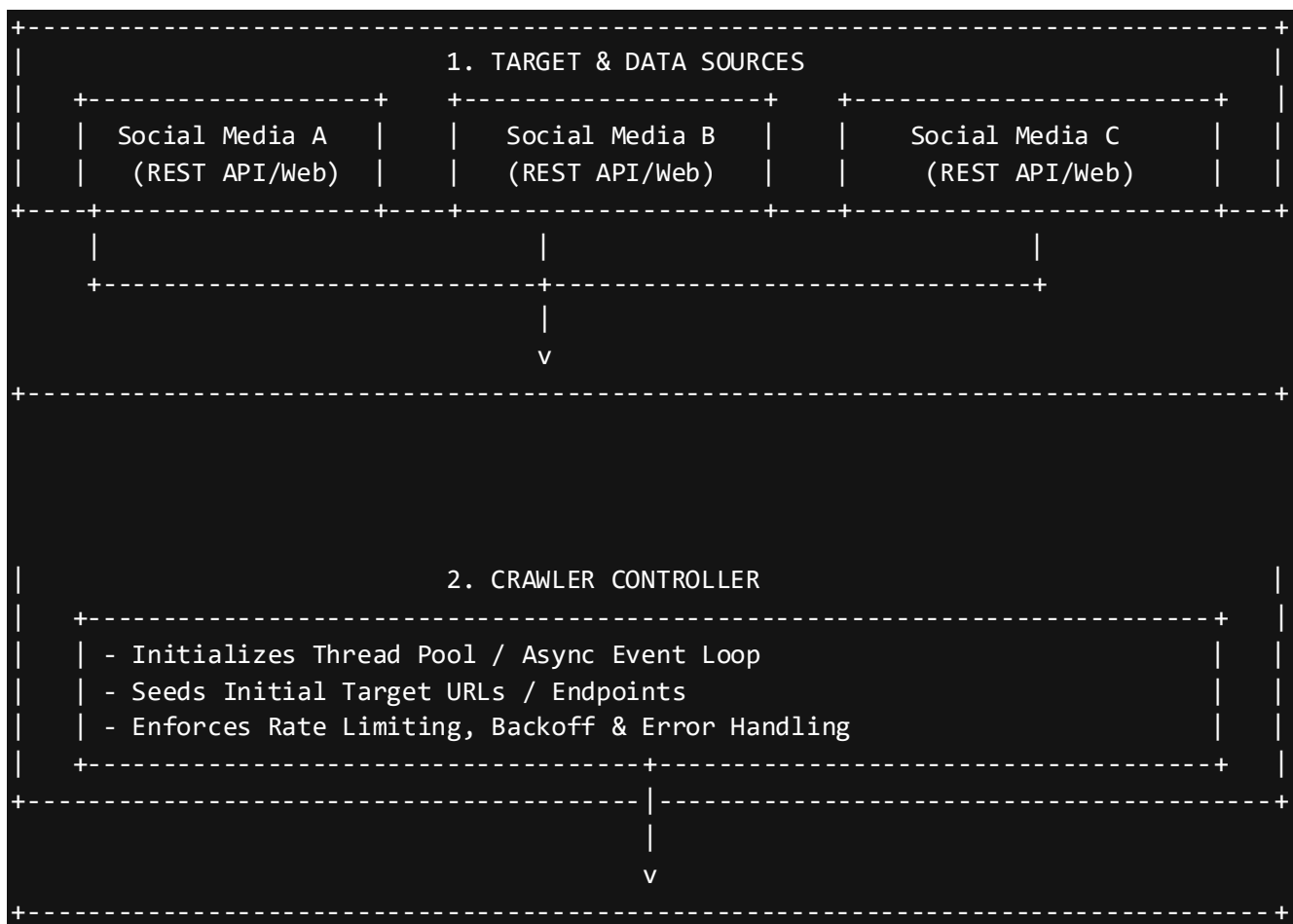
Subsystem Architecture Breakdown:

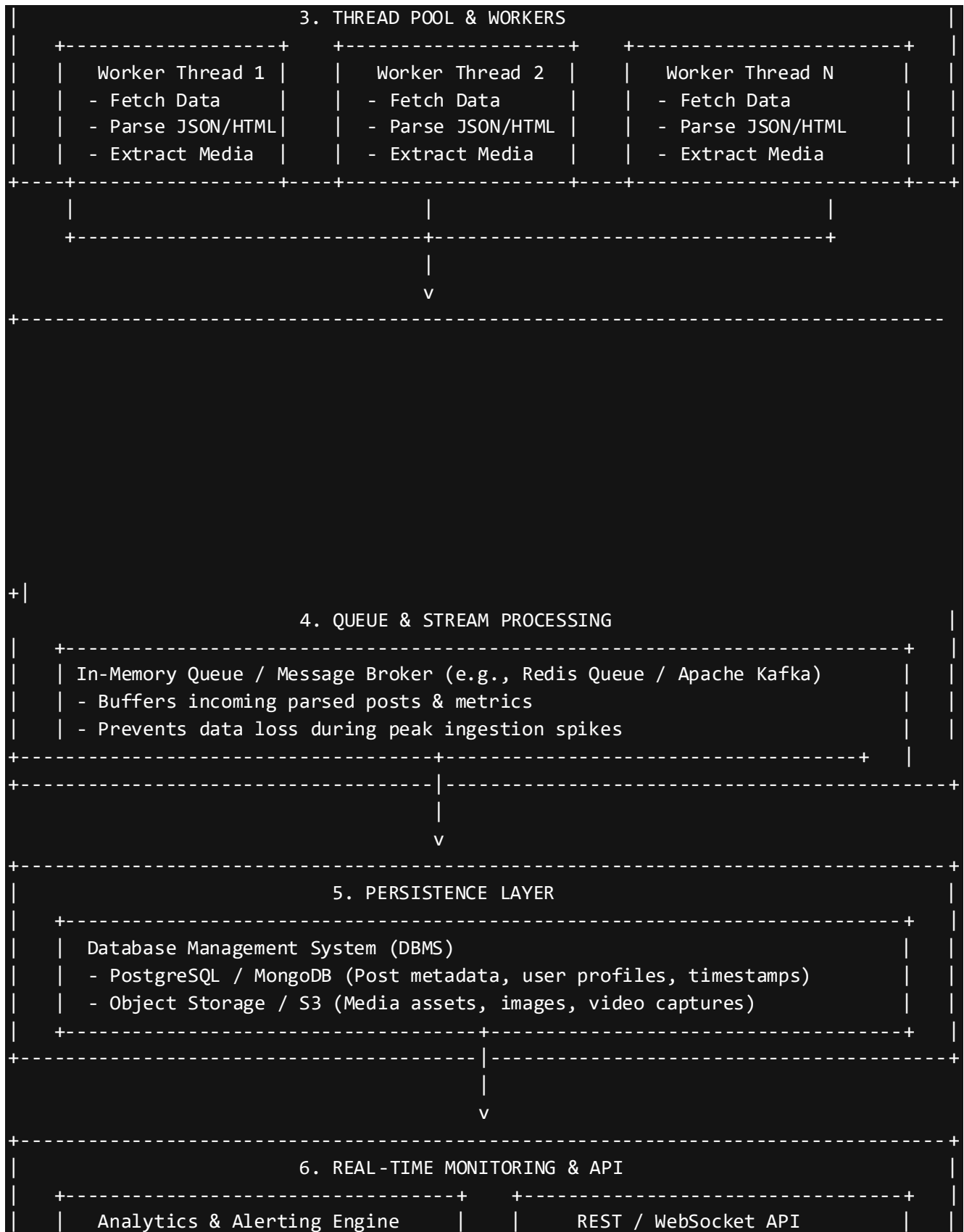
- **User Interface Layer:** Enables user registration, profile/interest setup, skill assessment execution, visual review of recommended career paths, skill roadmaps, and progress tracking.
- **Application Layer:** Contains three main operational modules: (1) User Profile & Skill Assessment Module, (2) AI Career Recommendation Module, and (3) Skill Gap & Roadmap Generation Module.
- **Data Layer:** Maintains persistent stores including User Database, Skills Database, Career Database, Courses & Resources DB, and Roadmap & Progress DB.

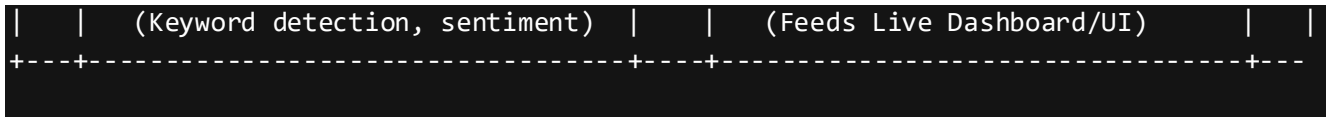
- **External Data Sources:** Integrates real-world signals from Online Course Platforms, Job Market/Industry Data, and Skill Ontologies/Taxonomies.
- **AI / Processing Engine:** Executes Machine Learning Models, Recommendation Engines, NLP Processing, and Skill Mapping & Analysis.
- **System Infrastructure:** Supported by Cloud Servers, Application Servers, Database Servers, Security & Authentication modules, and Backup & Recovery services.

2. Architecture Reconstruction

System Architecture Layout & Layer Interactions:







Module Interaction Flow:

- The User Interface Layer sends user inputs (skills, academic stage, stream, interests) to the Application Layer.
- Module 1 formats input data into structured user profiles and saves them into the User DB and Skills DB.
- Module 2 queries the Career DB and external Job Market Data, invoking the AI/Processing Engine (ML models and Similarity Matching) to evaluate candidate roles and return ranked career options.
- Module 3 compares the selected profile against role requirements stored in Skill Ontologies, determines gaps, retrieves recommendations from the Courses & Resources DB, and stores generated roadmaps in the Roadmap & Progress DB.

3. Algorithm Identification

The system combines rule-based symbolic reasoning with statistical NLP methods to handle structured and qualitative skill data:

- **Logical Reasoning & Rule-Based Agent Architecture:** Uses Forward Chaining to infer candidate career streams and high-growth job roles from known user attributes (e.g., Stream = Science, Duration = 2 Years → Modern High Growth Careers = AI/ML Engineering, Cybersecurity). Uses Backward Chaining to verify whether a user meets specific career prerequisites by tracing backwards from goal requirements. Employs Unification & Formal Logical Reasoning to match variable skill sets across non-standardized user submissions against predefined job taxonomies.
- **Natural Language Processing & Vector Matching:** Employs TF-IDF and Cosine Similarity / Similarity Matching to convert qualitative user text (interests, descriptions, strengths) into vector space representations, scoring alignment against job profiles.
- **Decision Logic:** Evaluates multi-attribute vectors (e.g., Academic Stream, High Growth Potential, Stability/Demand Rating) to output ranked recommendations.

4. Data Flow & Process Mapping

[User Input: Stream, Duration, Interests, Strengths]



[Data Preprocessing & NLP / TF-IDF Vectorization]



[Knowledge Base Lookup & Logical Inference Engine (Forward Chaining)]



[Ranked Career Recommendations (Filtered by Stability & Growth Rating)]



[Skill Gap Analysis (Target Skills minus Current Skills)]



[Roadmap & Resource Allocation (Mapping Courses, Certifications, Milestones)]



[Output Visualisation: Profile Dashboards & Progress Tracking]

5. Improvement & Redesign

To transition this prototype into a distributed, enterprise-grade system, the following architectural enhancements are proposed:

- **Scalability & Distributed Processing (PySpark & RDD Integration):** Transition batch dataset processing (e.g., multi-stream career spreadsheets) to Apache Spark (PySpark). Implement Resilient Distributed Datasets (RDDs) and Spark DataFrames to perform distributed skill-gap vector calculations across millions of candidate profiles simultaneously.
- **Real-Time Data Pipeline & RAG-LLM Hybrid Integration:** Replace static lookup tables with dynamic live job market feeds via continuous web scrapers and Kafka streaming queues. Integrate Retrieval-Augmented Generation (RAG) with Large Language Models (LLMs) alongside formal logical rules to handle ambiguous career queries with high explainability.

- **Probabilistic & Anomaly-Aware Reasoning:** Introduce Bayesian Networks / Probabilistic Reasoning into the inference engine to handle incomplete user inputs and dynamic market shifts. Incorporate automated assessment fraud/anomaly detection (e.g., Isolation Forest models) during skill evaluations.